

PRACTICAL 6: Parameterized Stored Procedure

AIM: Design a webpage to demonstrate the working of parameterized stored procedures.

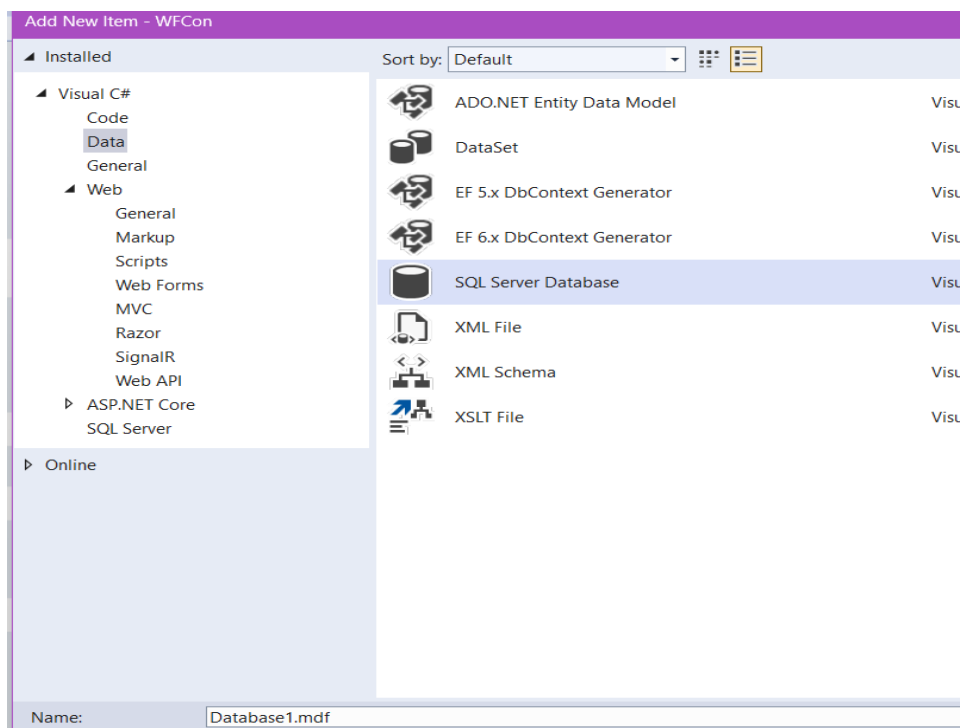
Step 1:

1. Open Visual Studio
2. Click **Create New Project**
3. Select **ASP.NET Web Application (.NET Framework)**
4. Click **Next**
5. Enter Project Name → ParameterizedProc
6. Click **Create**
7. Select **Empty Template**
8. Tick **Web Forms**
9. Click **Create**

DATABASE PART

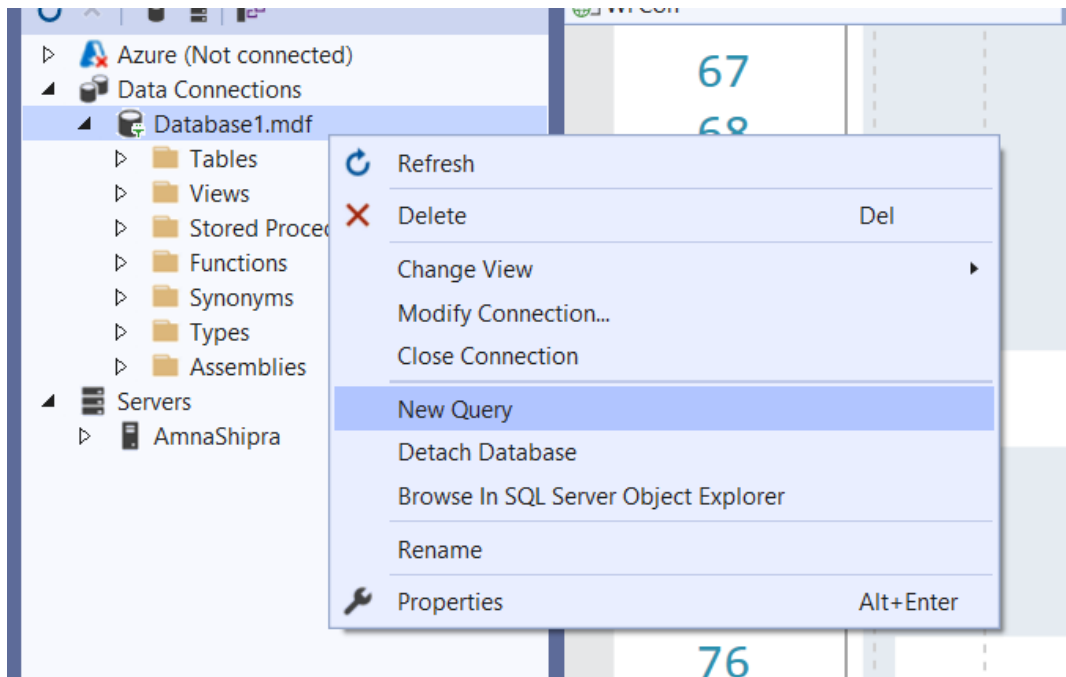
Step 2: Add Database

1. Right-click Project → **Add** → **New Item**
2. Select **SQL Server Database**
3. Name → Database1.mdf



Step 3: Create Table

Right Click on Database → New Query



Run this query:

```
CREATE TABLE UserRegistration  
(  
    C_Id INT IDENTITY(1,1) PRIMARY KEY,  
    C_Name VARCHAR(100),  
    C_Age VARCHAR(100),  
    C_Country VARCHAR(100)  
);
```

Step 4: Create Parameterized Stored Procedure

1. Again add new query:

```
CREATE PROCEDURE SpMyProcedure  
(  
    @Id INT = NULL,  
    @Name VARCHAR(100) = NULL,  
    @Age VARCHAR(100) = NULL,  
    @Country VARCHAR(100) = NULL,  
    @Action VARCHAR(100) = NULL
```

```

)
AS
BEGIN

    IF @Action = 'Insert'

        INSERT INTO UserRegistration(C_Name, C_Age, C_Country)

        VALUES(@Name, @Age, @Country)

    IF @Action = 'Update'

        UPDATE UserRegistration

        SET C_Name = @Name,

            C_Age = @Age,

            C_Country = @Country

        WHERE C_Id = @Id

    IF @Action = 'Delete'

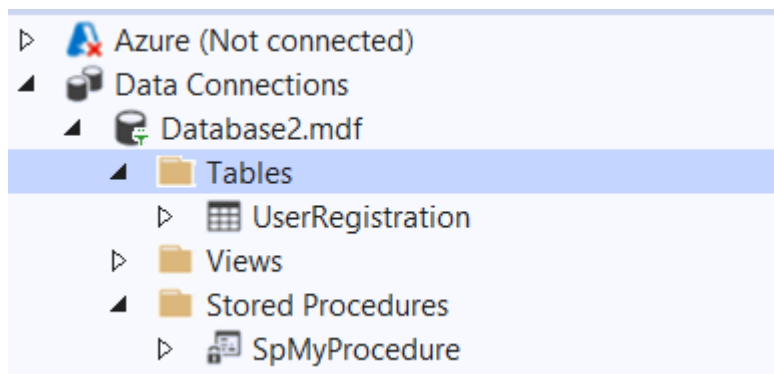
        DELETE FROM UserRegistration

        WHERE C_Id = @Id

END

```

2. Save the procedure



Step 6: Add Web Form

1. Right click Project → **Add** → **Web Form**
2. Name → WebForm1.aspx
3. Click Add

Step 7: Design UI (Frontend)

FRONTEND

WebForm1.aspx

```
<body>
    <form id="form1" runat="server">
        <asp:Label ID="Label1" runat="server" Text="ID"></asp:Label>
        <asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>
        <br /><br />

        <asp:Label ID="Label2" runat="server" Text="Name"></asp:Label>
        <asp:TextBox ID="TextBox2" runat="server"></asp:TextBox>
        <br /><br />

        <asp:Label ID="Label3" runat="server" Text="Age"></asp:Label>
        <asp:TextBox ID="TextBox3" runat="server"></asp:TextBox>
        <br /><br />

        <asp:Label ID="Label4" runat="server" Text="Country"></asp:Label>
        <asp:TextBox ID="TextBox4" runat="server"></asp:TextBox>
        <br /><br />

        <asp:Button ID="Button1" runat="server" Text="Show Data" OnClick="Button1_Click" />
        &nbsp;
        <asp:Button ID="Button2" runat="server" Text="Insert" OnClick="Button2_Click" />
        &nbsp;
        <asp:Button ID="Button3" runat="server" Text="Update" OnClick="Button3_Click" />
        &nbsp;
        <asp:Button ID="Button4" runat="server" Text="Delete" OnClick="Button4_Click" />
        &nbsp;
        <asp:Button ID="Button5" runat="server" Text="Clear" OnClick="Button5_Click" />

    <br /><br />

    <asp:GridView ID="GridView1" runat="server"></asp:GridView>
    </form>
</body>
```

The screenshot shows the design view of the WebForm1.aspx page. It features a form with four text boxes labeled "ID", "Name", "Age", and "Country". Below the text boxes are five buttons: "Show Data", "Insert", "Update", "Delete", and "Clear". At the bottom of the form is a GridView control displaying a table with 5 rows and 3 columns. The columns are labeled "Column0", "Column1", and "Column2". The data in the GridView is as follows:

Column0	Column1	Column2
abc	abc	abc
abc	abc	abc
abc	abc	abc
abc	abc	abc
abc	abc	abc

Backend Code:

WebForm1.aspx.cs

```
using System;
using System.Data;
using System.Data.SqlClient;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace ParameterizedProc
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        string str = @"Data Source=(LocalDB)\MSSQLLocalDB;
                      AttachDbFilename=|DataDirectory|\Database2.mdf;
                      Integrated Security=True";

        // SHOW DATA
        protected void Button1_Click(object sender, EventArgs e)
        {
            SqlConnection con = new SqlConnection(str);

            SqlDataAdapter da = new SqlDataAdapter("SELECT * FROM UserRegistration",
con);

            DataSet ds = new DataSet();

            da.Fill(ds);

            GridView1.DataSource = ds;
            GridView1.DataBind();
        }

        // INSERT
        protected void Button2_Click(object sender, EventArgs e)
        {
            SqlConnection con = new SqlConnection(str);

            SqlCommand cmd = new SqlCommand("SpMyProcedure", con);
            cmd.CommandType = CommandType.StoredProcedure;

            cmd.Parameters.AddWithValue("@Action", "Insert");
            cmd.Parameters.AddWithValue("@Name", TextBox2.Text);
            cmd.Parameters.AddWithValue("@Age", TextBox3.Text);
            cmd.Parameters.AddWithValue("@Country", TextBox4.Text);

            con.Open();
            cmd.ExecuteNonQuery();
            con.Close();

            Response.Write("Inserted Successfully");
        }

        // UPDATE
        protected void Button3_Click(object sender, EventArgs e)
        {
            SqlConnection con = new SqlConnection(str);

            SqlCommand cmd = new SqlCommand("SpMyProcedure", con);
            cmd.CommandType = CommandType.StoredProcedure;

            cmd.Parameters.AddWithValue("@Action", "Update");
            cmd.Parameters.AddWithValue("@Id", TextBox1.Text);
        }
    }
}
```

```

        cmd.Parameters.AddWithValue("@Name", TextBox2.Text);
        cmd.Parameters.AddWithValue("@Age", TextBox3.Text);
        cmd.Parameters.AddWithValue("@Country", TextBox4.Text);

        con.Open();
        cmd.ExecuteNonQuery();
        con.Close();

        Response.Write("Updated Successfully");
    }

    // DELETE
    protected void Button4_Click(object sender, EventArgs e)
    {
        SqlConnection con = new SqlConnection(str);

        SqlCommand cmd = new SqlCommand("SpMyProcedure", con);
        cmd.CommandType = CommandType.StoredProcedure;

        cmd.Parameters.AddWithValue("@Action", "Delete");
        cmd.Parameters.AddWithValue("@Id", TextBox1.Text);

        con.Open();
        cmd.ExecuteNonQuery();
        con.Close();

        Response.Write("Deleted Successfully");
    }

    // CLEAR
    protected void Button5_Click(object sender, EventArgs e)
    {
        TextBox1.Text = "";
        TextBox2.Text = "";
        TextBox3.Text = "";
        TextBox4.Text = "";
    }
}

```

ID

Name

Age

Country

Show Data

Insert

Update

Delete

Clear

C_Id	C_Name	C_Age	C_Country
2	Abc	25	qwerty
4	Amna	25	amabnba
5	asdfghj	20	ojbggb
6	efffr	22	asdf